homework12

P364-习题 12.6	A 类	2-1, 2-3, 2-9; 3;
	B 类	3
P380-习题 12.7	A 类	4-4; 5;

2-1、幂级数展开 $\dfrac{1+x}{(1-x)^2}$

解: 由

$$\frac{1}{1-x}=\sum_{n\geq 0}x^n \Rightarrow \frac{1}{(1-x)^2}=\sum_{n\geq 1}n\,x^{n-1}=\sum_{n\geq 0}(n+1)x^n$$

$$\Rightarrow \frac{1+x}{(1-x)^2}=\frac{2}{(1-x)^2}-\frac{1}{1-x}=\sum_{n\geq 0}(2n+1)x^n$$

由$\dfrac{2n+3}{2n+1}\to 1 \Rightarrow R=1$ $x=\pm 1$时$\sum_{n\geq 0}(2n+1)x^n$发散 $\Rightarrow$ 收敛域为 $(-1,1)$

2-3、幂级数展开 $\ln\dfrac{1+x}{1-x}$

解: 由

$$\big(\ln(1+x)\big)' = \frac{1}{1+x} = \sum_{n\geq 0}(-1)^n x^n$$

$$\Rightarrow \ln(1+x) = \sum_{n\geq 0}\frac{(-1)^n}{n+1}x^{n+1} = \sum_{n\geq 1}\frac{(-1)^{n-1}}{n}x^n \Rightarrow \ln(1-x) = \sum_{n\geq 1}\frac{-1}{n}x^n$$

$$\Rightarrow \ln\frac{1+x}{1-x} = \sum_{n\geq 1}\frac{(-1)^{n-1}+1}{n}x^n = \sum_{m\geq 0}\frac{2}{2m+1}x^{2m+1}, \quad x\in(-1,1)$$

2-9、幂级数展开 e^{-x^2}

解: 由

$$e^z = \sum_{n\geq 0}\frac{z^n}{n!} \quad z\in\mathbb{R} \Rightarrow e^{-x^2} = \sum_{n\geq 0}\frac{(-1)^n}{n!}x^{2n} \quad x\in\mathbb{R}$$

3、幂级数展开 $f(x)=\ln(5+x)$

解: 由

$$f'(x)=\frac{1}{5+x}=\frac{1}{5}\frac{1}{1+\frac{x}{5}}=\frac{1}{5}\sum_{n\geq 0}\frac{(-1)^n}{5^n}x^n$$

$$\Rightarrow f(x)=c+\sum_{n\geq 0}\frac{(-1)^n}{5^{n+1}(n+1)}x^{n+1},\ f(0)=\ln 5$$

$$\Rightarrow f(x)=\ln 5+\sum_{n\geq 1}\frac{(-1)^{n-1}}{n\,5^n}x^n,\quad x\in(-5,5]$$

3、展开 $\arcsin x$, 并求 $\sum_{n=1}^{\infty} \frac{1}{n^2}$

解: 记 $f(x) = \arcsin x$, 则 $f'(x) = \frac{1}{\sqrt{1-x^2}}$, 由

$$\frac{1}{\sqrt{1-t}} = 1 + \sum_{n\geq 1} \frac{(2n-1)!!}{2^n n!} t^n \Rightarrow f'(x) = 1 + \sum_{n\geq 1} \frac{(2n-1)!!}{2^n n!} x^{2n}, f(0) = 0$$

$$\Rightarrow f(x) = x + \sum_{n\geq 1} \frac{(2n-1)!!}{2^n n!(2n+1)} x^{2n+1} = \sum_{n\geq 0} \frac{(2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1} \quad x \in [-1, 1]$$

...

4-4、傅里叶展开 $f(x) = |\sin x|$

解: 依题意 $b_n = 0$,

$$a_0 = \frac{2}{\pi}\int_0^{\pi} \sin x\, dx = \frac{2}{\pi}(-\cos x)\Big|_0^{\pi} = \frac{4}{\pi}$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} \sin x \cos nx\, dx = \frac{1}{\pi}\int_0^{\pi} \big(\sin(n+1)x - \sin(n-1)x\big)\, dx$$

$$\Rightarrow a_1 = \frac{1}{\pi}\int_0^{\pi} \sin 2x = \frac{\cos 2x}{-\pi}\Big|_0^{\pi} = 0$$

$$a_n = \frac{1}{\pi}\left(\frac{\cos(n+1)x}{-(n+1)} + \frac{\cos(n-1)x}{n-1}\right)\Big|_0^{\pi} = \frac{1}{\pi}\left(\frac{(-1)^n}{n+1} + \frac{(-1)^{n-1}}{n-1}\right) = \frac{2}{\pi}\frac{(-1)^{n-1}}{n^2-1}$$

$$\Rightarrow S(x) = \frac{2}{\pi} + \sum_{n\geq 2} \frac{2(-1)^{n-1}}{\pi(n^2-1)} \cos\, nx = |\sin x| \quad x \in (-\pi, \pi]$$

5、傅里叶展开 $f(x)=\begin{cases}x^2 & 0<x<\pi\\ 0 & x=\pi\\ -x^2 & \pi<x\le 2\pi\end{cases}$，并求级数 $\sum\limits_{n\ge 1}\dfrac{1}{n^2}$

解：依题意，

$$a_0=\frac{1}{\pi}\int_0^{2\pi}f(x)\,dx=\frac{1}{\pi}\Big(\int_0^{\pi}x^2\,dx-\int_{\pi}^{2\pi}x^2\,dx\Big)=\frac{1}{\pi}\Big(\frac{x^3}{3}\Big|_0^{\pi}-\frac{1}{3}x^3\Big|_{\pi}^{2\pi}\Big)=-2\pi^2$$

$$a_n=\frac{1}{\pi}\int_0^{2\pi}f(x)\cos nx\,dx=\frac{1}{\pi}\Big(\int_0^{\pi}x^2\cos nx\,dx-\int_{\pi}^{2\pi}x^2\cos nx\,dx\Big)$$

$$\int x^2\cos nx\,dx=\ldots=\frac{x^2}{n}\sin nx+\frac{2x}{n^2}\cos nx-\frac{2}{n^3}\sin nx+c=F(x)+c$$

$$\Rightarrow a_n=\frac{1}{\pi}\big(F(x)\big|_0^{\pi}-F(x)\big|_{\pi}^{2\pi}\big)=\frac{1}{\pi}\big(\frac{2\pi}{n^2}(-1)^n-\frac{4\pi}{n^2}+\frac{2\pi}{n^2}(-1)^n\big)=\frac{4}{n^2}\big((-1)^n-1\big)$$

$$\int x^2 \sin nx\,dx = \ldots = \frac{x^2}{-n}\cos nx + \frac{2x}{n^2}\sin nx + \frac{2}{n^3}\cos nx + c = G(x) + c$$

$$\begin{aligned} b_n &= \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\left(\int_0^{\pi} x^2\sin nx\,dx - \int_{\pi}^{2\pi} x^2\sin nx\,dx\right) \\ &= \frac{1}{\pi}\left(G(x)\Big|_0^{\pi} - G(x)\Big|_{\pi}^{2\pi}\right) = \frac{1}{\pi}\big(2G(\pi) - G(0) - G(2\pi)\big) \\ &= \frac{1}{\pi}\left(2(-1)^n\left(\frac{2}{n^3} - \frac{\pi^2}{n}\right) - \frac{\pi^2}{n}\right) \end{aligned}$$

$$\Rightarrow S(x) = -\pi^2 + \sum_{m\geq 0}\frac{-8}{(2m+1)^2}\cos(2m+1)x$$

$$+\sum_{n\geq 1}\frac{1}{\pi}\left(2(-1)^n\left(\frac{2}{n^3} - \frac{\pi^2}{n}\right) - \frac{\pi^2}{n}\right)\sin nx = \begin{cases} x^2 & x\in(0,\pi) \\ -x^2 & x\in(\pi,2\pi) \\ 0 & x=\pi \\ -2\pi^2 & x=2\pi. \end{cases}$$

$$\text{取}x=0 \Rightarrow -\pi^2 - 8\sum_{m\geq 0}\frac{1}{(2m+1)^2} = -2\pi^2$$

$$\Rightarrow \sum_{m\geq 0}\frac{1}{(2m+1)^2} = \frac{\pi^2}{8} = I_{\text{奇}} = I_0 - I_{\text{偶}} = I_0 - \frac{1}{4}I_0 \Rightarrow I_0 = \frac{\pi^2}{6}$$